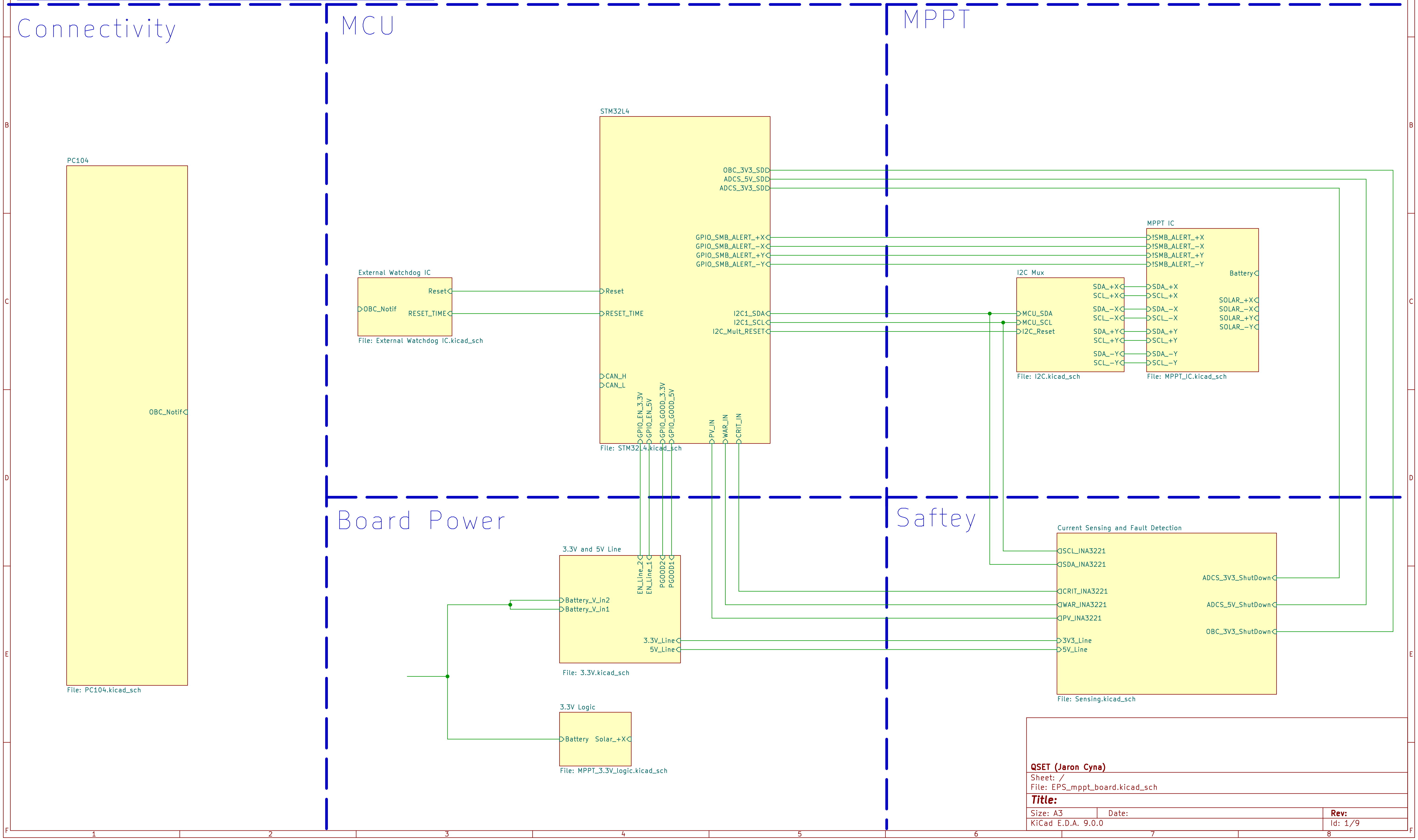
Overview

This is the second QSET EPS board for the mission planned for launch in 2028–2029. It was designed by Jaron Cyna (23wbjx@queensu.ca). Feel free to reach out to me if there are any questions regarding the board.

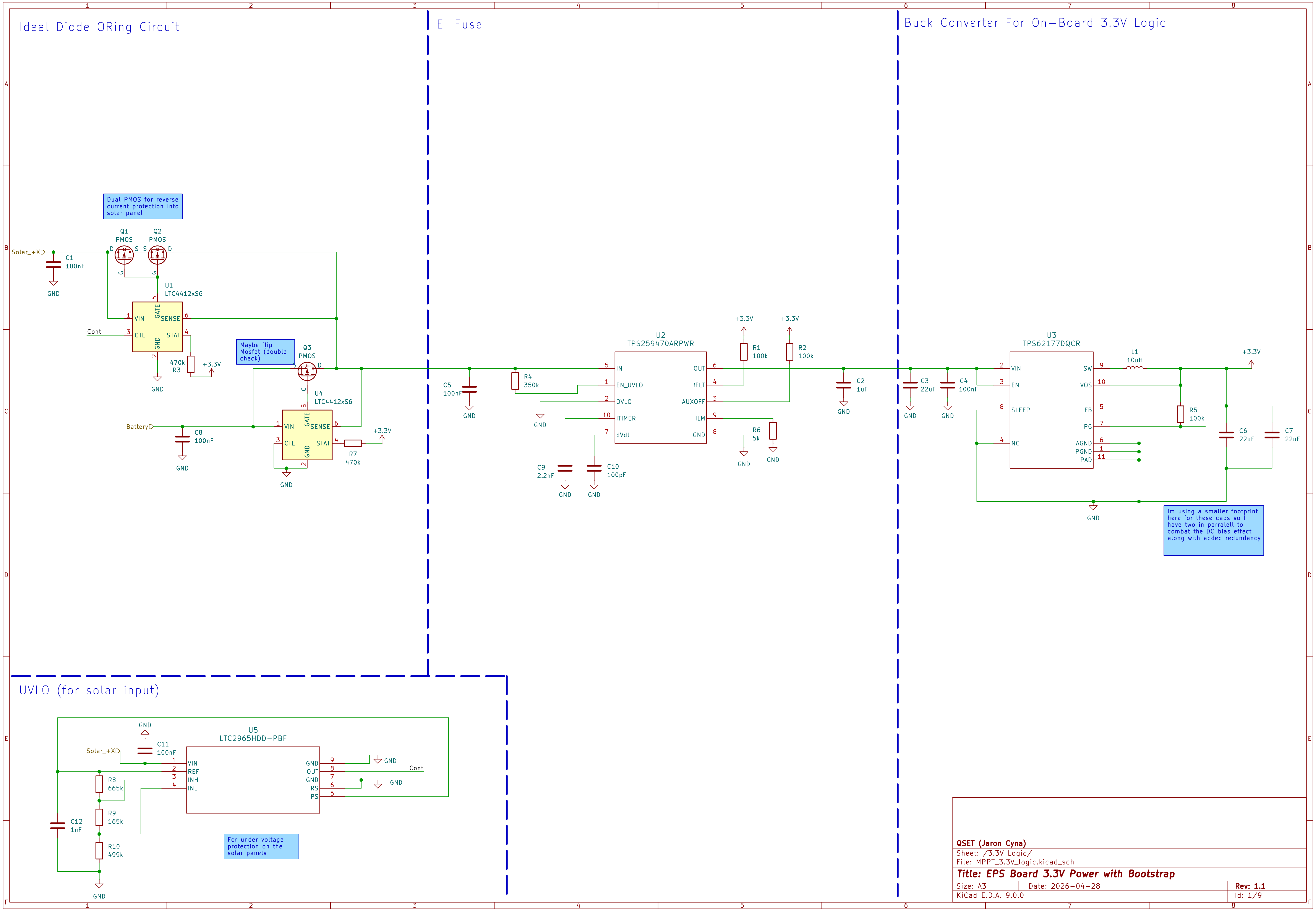
HIGH LEVEL BOARD VIEW:
This board is meant to provide a few key properties to the board:
1: Maximum Power Point Tracking (MPPT) to track and maintain optimal charging speed from solar panels
2: 3V3 and 5V lines to provide power to ADCS, OBC, and (possibly) comms
3: Telemetry and fault detection which is processed by the onboard MCU and sent to OBC for interpretation over CAN
    3.1: Telemetry includes many integrated elements of the different IC's, external MCU watchdog, and a current sensor on all lines.
    3.2: Fault detection also has integrated elements from the ICs, but the primary detection comes from the E–Fuses integrated throughout the board

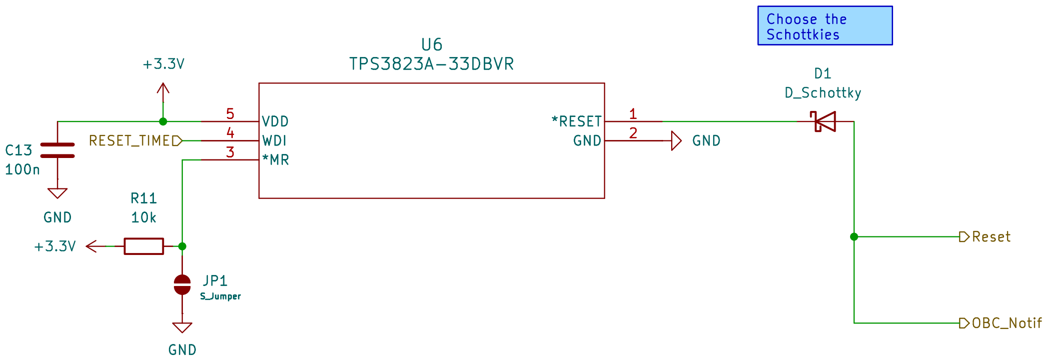



**QSET (Jaron Cyna)**
Sheet: /
File: EPS_mppt_board.kicad_sch

***Title:***

Size: A3 | Date: | **Rev:**
KiCad E.D.A. 9.0.0 | Id: 1/9


Ideal Diode ORing Circuit
E-Fuse
Buck Converter For On-Board 3.3V Logic
Dual PMOS for reverse current protection into solar panel
Solar_+XD
C1
100nF
GND
Q1
PMOS
Q2
PMOS
U1
LTC4412xS6
VIN
SENSE
CTL
STAT
GATE
GND
Cont
470k
R3
+3.3V
Maybe flip Mosfet (double check)
Q3
PMOS
U4
LTC4412xS6
BatteryD
C8
100nF
R7
470k
C5
100nF
R4
350k
U2
TPS259470ARPWR
IN
EN_UVLO
OVLO
ITIMER
dVdt
OUT
!FLT
AUXOFF
ILM
C9
2.2nF
C10
100pF
R1
100k
R2
100k
R6
5k
C2
1uF
C3
22uF
C4
100nF
U3
TPS62177DQCR
VIN
EN
SLEEP
NC
SW
VOS
FB
PG
AGND
PGND
PAD
L1
10uH
R5
100k
C6
22uF
C7
22uF
Im using a smaller footprint here for these caps so I have two in parralell to combat the DC bias effect along with added redundancy
UVLO (for solar input)
U5
LTC2965HDD-PBF
C11
100nF
REF
INH
INL
OUT
RS
PS
R8
665k
R9
165k
R10
499k
C12
1nF
For under voltage protection on the solar panels
QSET (Jaron Cyna)
Sheet: /3.3V Logic/
File: MPPT_3.3V_logic.kicad_sch
Title: EPS Board 3.3V Power with Bootstrap
Size: A3
Date: 2026-04-28
Rev: 1.1
KiCad E.D.A. 9.0.0
Id: 1/9



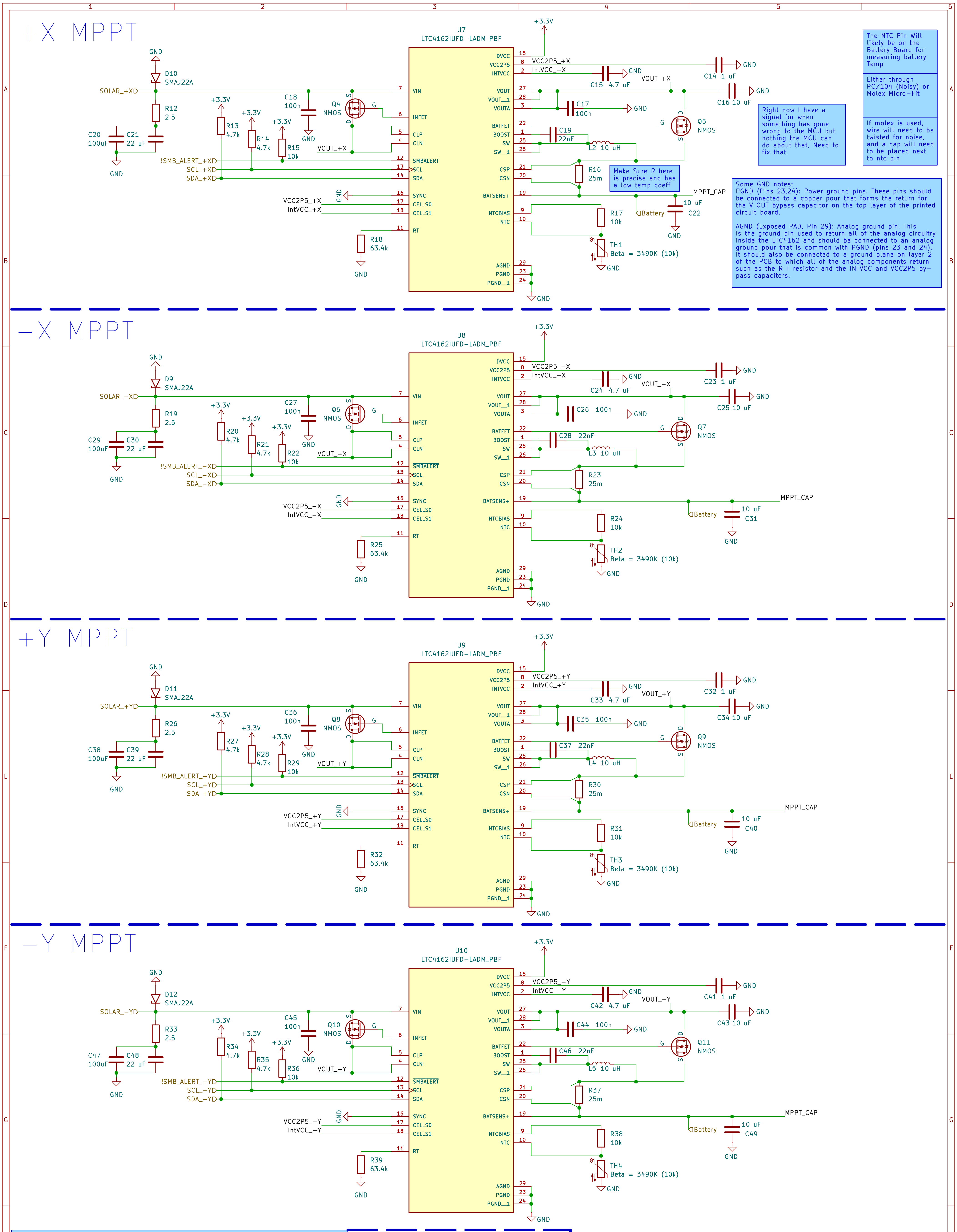

**QSET (Jaron Cyna)**

Sheet: /External Watchdog IC/

File: External Watchdog IC.kicad_sch

## Title: External Watchdog

| | | |
|---|---|---|
| Size: A4 | Date: 2026-03-17 | **Rev: 1.0** |
| KiCad E.D.A. 9.0.0 | | Id: 2/9 |




Notes:
-!SMB_ALERT will be tied to a External Interrupt Line (EXTI) on the MCU which will be triggered by a falling edge
- Think about ferrite Bead Placement on input
- Assumptions made when calculating inductor val:
15V solar input, 8.4V battery, 1.2A max charge limit
10uH (max I= 3A) (<30m ohms)

Switched to IC based MPPT from a discrete one for Power and Reliabillity

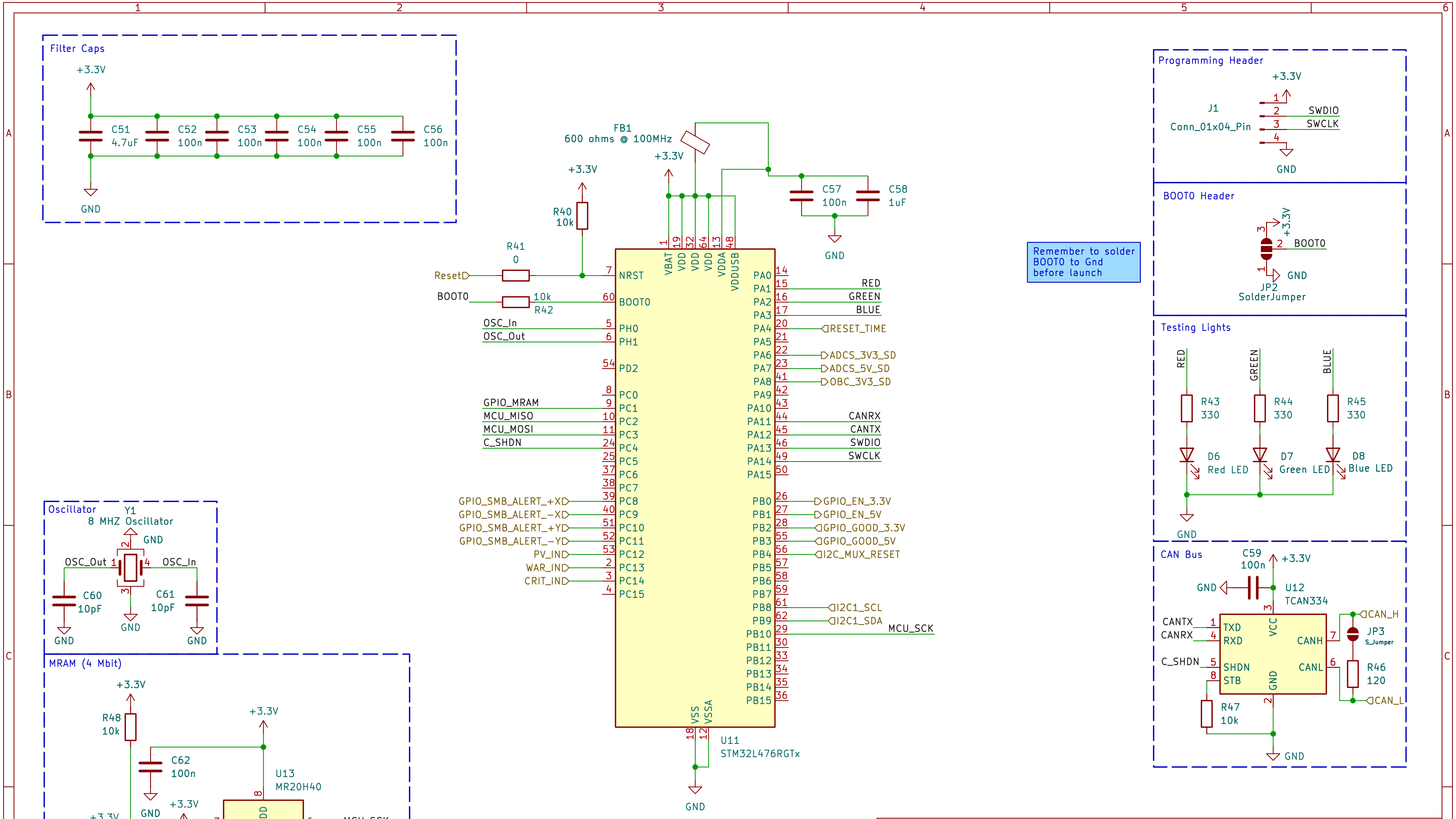
QSET (Jaron Cyna)

Sheet: /MPPT IC/

File: MPPT_IC.kicad_sch

**Title: MPPT**

| | | |
|---|---|---|
| Size: A3 | Date: 2026-03-17 | Rev: 1.1 |
| KiCad E.D.A. 9.0.0 | | Id: 3/9 |



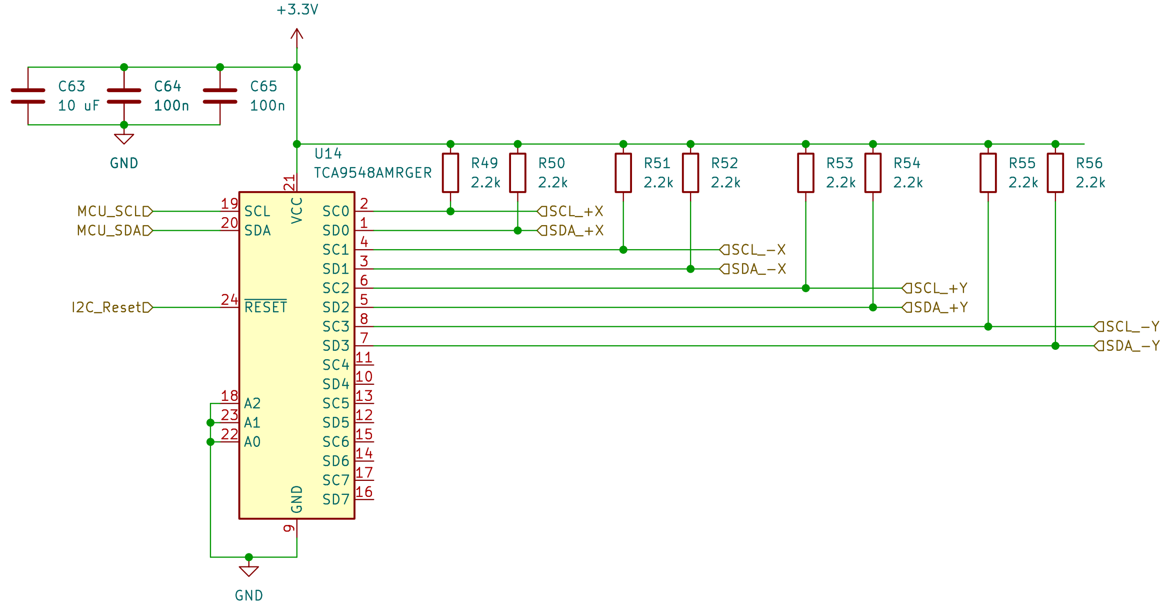

Switched to STM32L4 from G4

**QSET (Jaron Cyna)**

Sheet: /STM32L4/
File: STM32L4.kicad_sch

***Title: STM32L4 Schematic***

Size: A4 | Date: 2026-03-17 | **Rev: 1.1**
KiCad E.D.A. 9.0.0 | Id: 4/9



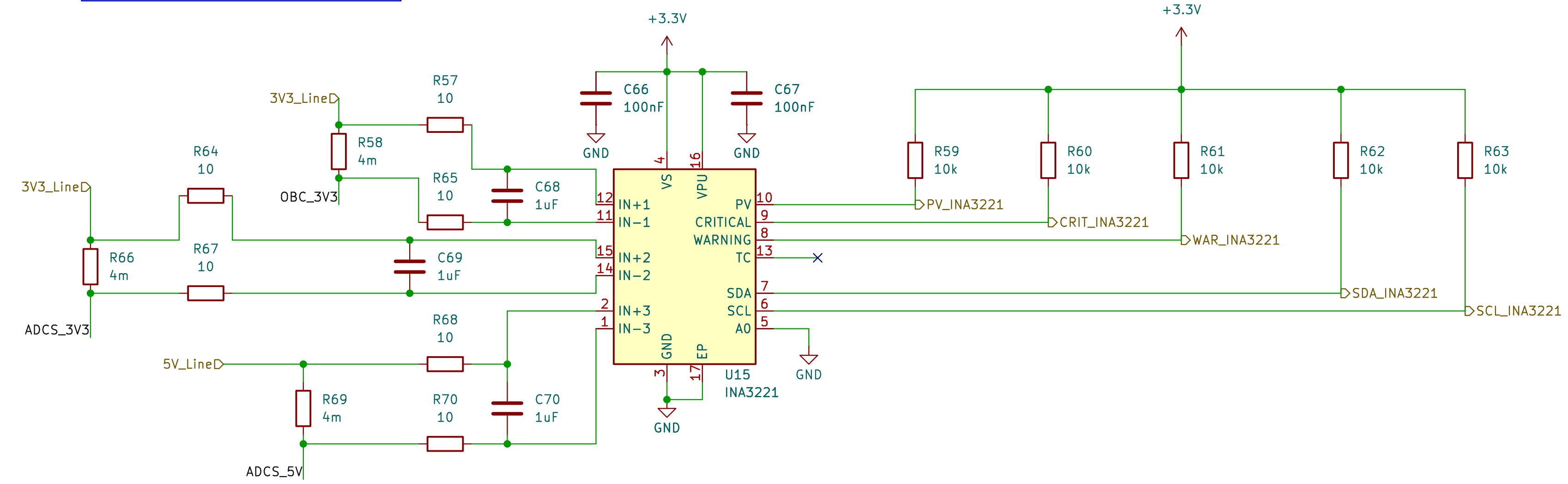

Notes:
Place decoupling Caps (0.1 uF) close to VCC and GND and 10uF cap still close but not as close

**QSET (Jaron Cyna)**
Sheet: /I2C Mux/
File: I2C.kicad_sch
***Title: I2C Multiplexer***
Size: A4 Date: 2026-03-17 **Rev: 1.0**
KiCad E.D.A. 9.0.0 Id: 5/9

# Current Sensing

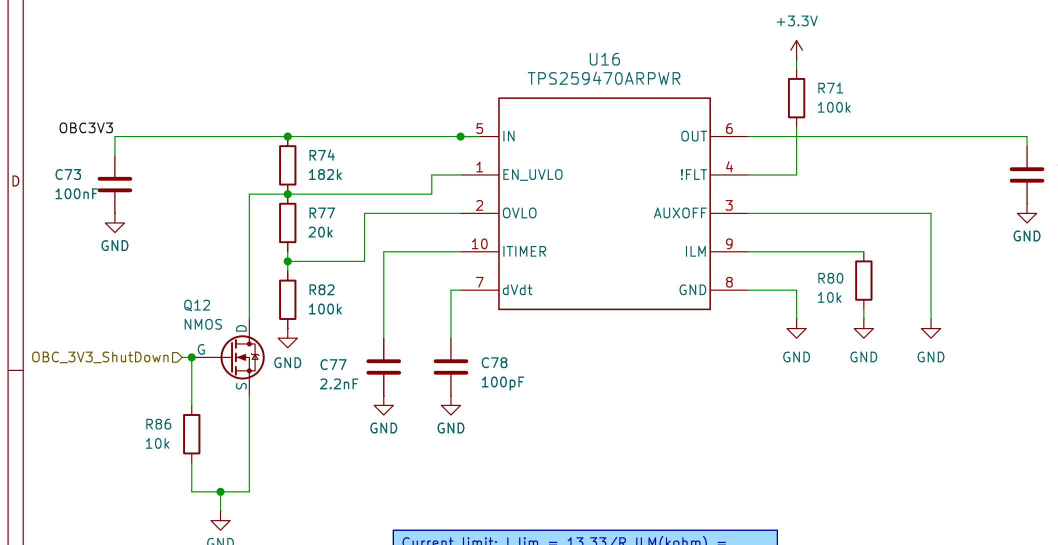
By using 4m ohm shunts, the equation
I_RES =  40*10^-6/R_SHUNT =
40*10^-6/0.004 = 0.01 A = 10mA

For the cutoff frequency of the RC filter,
the equation is f_c = 1/(2*pi*R*C) =
1/(2*pi*10*1*10^-6) = 16kHz




Use Kelvin Connections between 10 and 4m ohm resistors

# OBC 3.3V E−Fuse



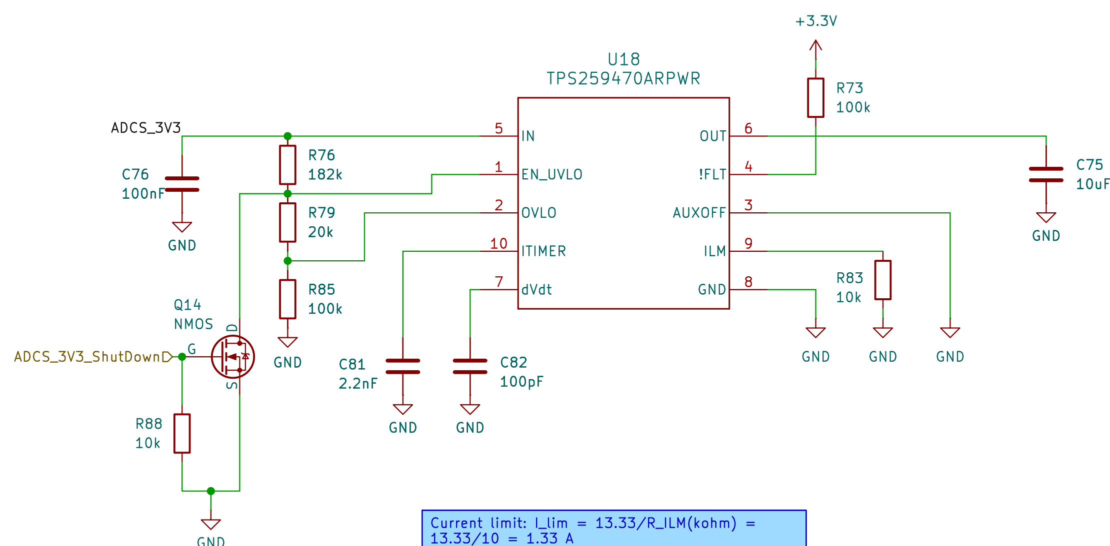

Current limit: I_lim = 13.33/R_ILM(kohm) = 13.33/10 = 1.33 A

Output Slew rate(dV/dt = 500/C_(dV/dt)(pF)) = 500/100 = 5 V/ms

for the Itimer:: t = C_ITIMER*0.2 = 2.2*0.2 = 0.44ms

For OVLO and UVLO, this will turn on when the voltage is above 3.02V and below 3.63V

# ADCS 3.3V E−Fuse



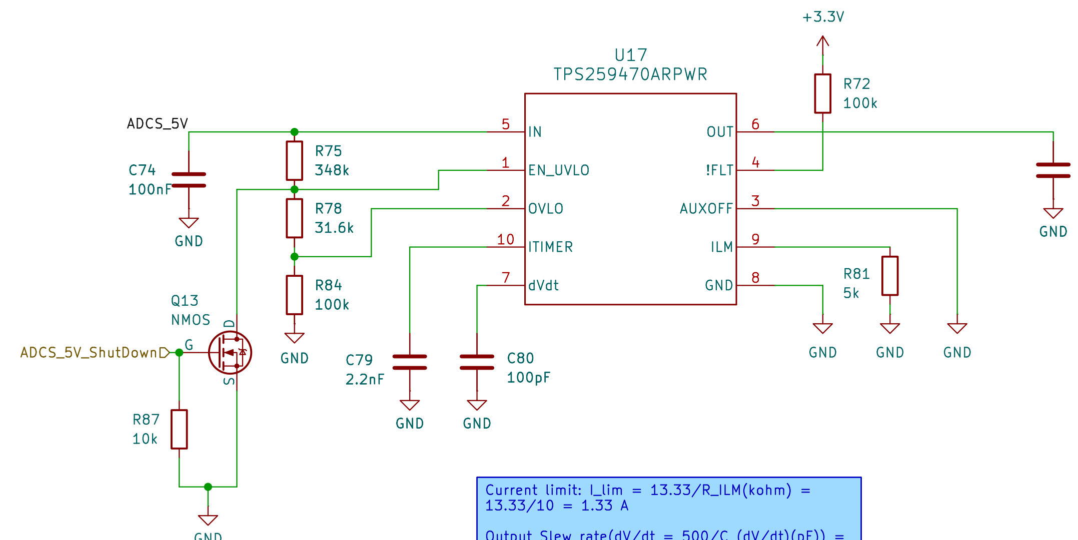

Current limit: I_lim = 13.33/R_ILM(kohm) = 13.33/10 = 1.33 A

Output Slew rate(dV/dt = 500/C_(dV/dt)(pF)) = 500/100 = 5 V/ms

for the Itimer:: t = C_ITIMER*0.2 = 2.2*0.2 = 0.44ms

For OVLO and UVLO, this will turn on when the voltage is above 3.02V and below 3.63V

# ADCS 5V E−Fuse




Current limit: I_lim = 13.33/R_ILM(kohm) = 13.33/10 = 1.33 A

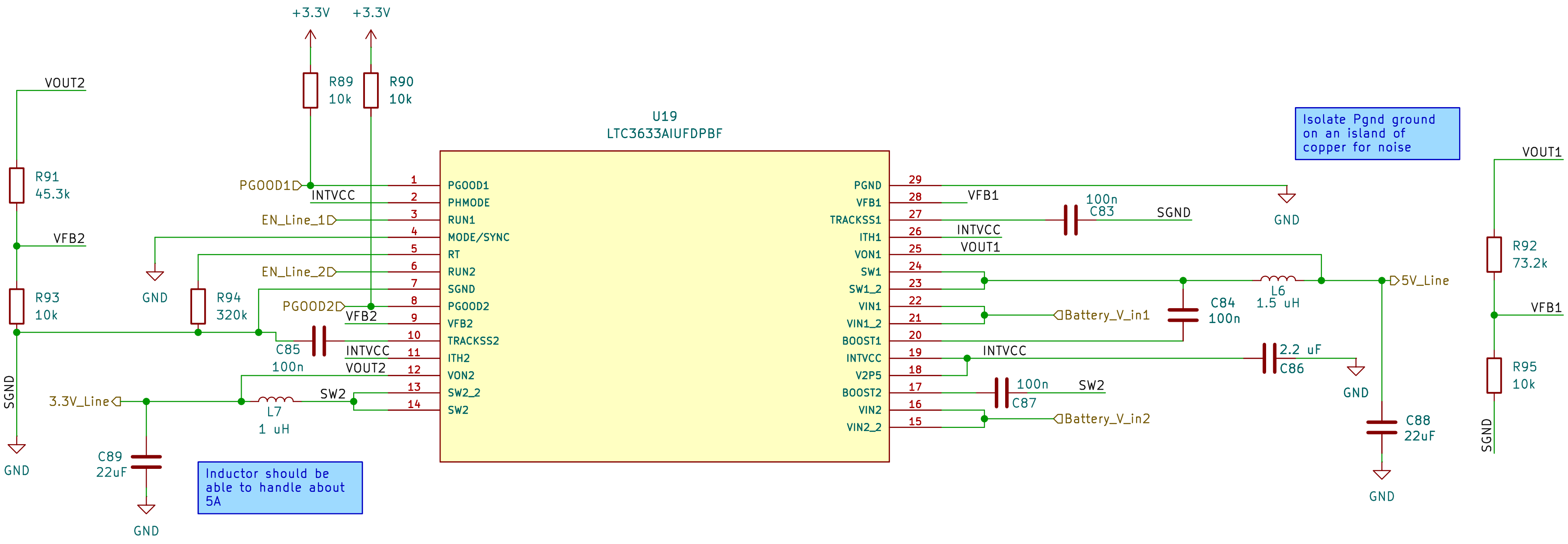
Output Slew rate(dV/dt = 500/C_(dV/dt)(pF)) = 500/100 = 5 V/ms

for the Itimer:: t = C_ITIMER*0.2 = 2.2*0.2 = 0.44ms

For OVLO and UVLO, this will turn on when the voltage is above 4.41V and below 5.85V

At 1A per rail, hypothetically, the power loss of this circuit would be 32mW per line, which would be less than a percent of a 3.3V 1A or 5V 1A power.

**QSET (Jaron Cyna)**
Sheet: /Current Sensing and Fault Detection/
File: Sensing.kicad_sch
***Title: Current Sensing and E−Fuse***
Size: User | Date: 2026−04−29 | **Rev: 1.0**
KiCad E.D.A. 9.0.0 | Id: 6/9




SGND and all signal must have a ground island to isolate from noise

Notes:
- Both the 3V and the 5V lines can supply 3A
- Use thick traces for anything carryiing high current
- Use copper islands for anything which is noisy or needs to be isolated from noise

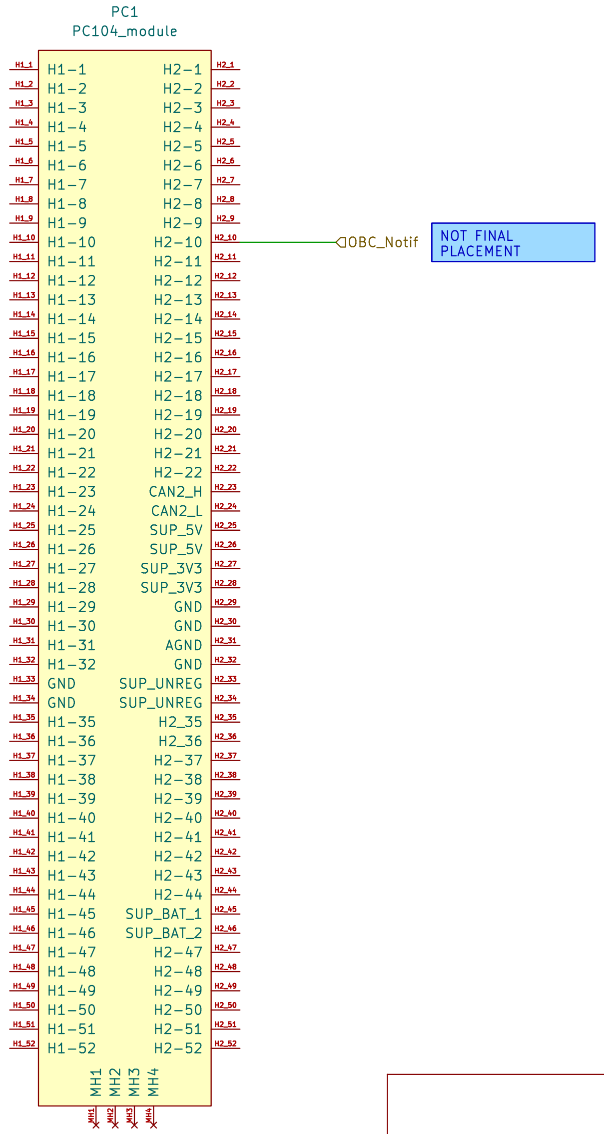
**QSET (Jaron Cyna)**

Sheet: /3.3V and 5V Line/
File: 3.3V.kicad_sch

***Title: 3.3V and 5V Buck Converter***

Size: A4 | Date: 2026-03-17 | **Rev: 1.0**

KiCad E.D.A. 9.0.0 | Id: 7/9




**QSET (Jaron Cyna)**

Sheet: /PC104/

File: PC104.kicad_sch

***Title:***

| | | |
|---|---|---|
| Size: A4 | Date: | **Rev:** |
| KiCad E.D.A. 9.0.0 | | Id: 8/9 |